

Separable Differential Equations

$$\boxed{1} \quad y' = 2xy$$

$$\int \frac{dy}{y} = \int 2x \, dx$$

$$\ln |y| = x^2 + C$$

$$y = Ce^{x^2}$$

$$\boxed{2} \quad (1 + x^2)y' = 1 + y^2$$

$$\int \frac{dy}{1 + y^2} = \int \frac{dx}{1 + x^2}$$

$$\arctan y = \arctan x + C$$

$$y = \tan(\arctan x + C)$$

$$\boxed{3} \quad y' = y \cos x$$

$$\int \frac{dy}{y} = \int \cos x \, dx$$

$$\ln |y| = \sin x + C$$

$$y = Ce^{\sin x}$$

$$\boxed{4} \quad y' = e^{x+y}$$

$$y' = e^x \cdot e^y$$

$$\int e^{-y} \, dy = \int e^x \, dx$$

$$-e^{-y} = e^x + C$$

$$e^{-y} = -e^x - C$$

$$y = -\ln(-e^x - C)$$

$$\boxed{5} \quad xy' = y \ln y$$

$$\int \frac{dy}{y \ln y} = \int \frac{dx}{x}$$

$$\ln |\ln y| = \ln |x| + C$$

$$\ln y = Cx$$

$$y = e^{Cx}$$

$$\boxed{6} \quad y' = y \ln x$$

$$\int \frac{dy}{y} = \int \ln x \, dx$$

$$u = \ln x, \quad dv = dx, \quad du = \frac{dx}{x}, \quad v = x$$

$$\int \ln x \, dx = x \ln x - \int x \cdot \frac{dx}{x} = x \ln x - x + C$$

$$\ln |y| = x \ln x - x + C$$

$$y = Ce^{x \ln x - x}$$

Homework

$$\boxed{1} \quad y' = 3x^2y$$

$$\boxed{2} \quad y' = (y + 1) \tan x$$

$$\boxed{3} \quad y' = xy^2$$

$$\boxed{4} \quad (1 + x)y \, dy = (1 - y)x \, dx$$

$$\boxed{5} \quad y' = e^{2x-y}$$

$$\boxed{6} \quad y' = y \tan x$$

$$\boxed{7} \quad y' = y^2 \cos x$$

$$\boxed{8} \quad xy \, dx + (x^2 + 1) \, dy = 0$$

$$\boxed{9} \quad y' \ln x = \frac{y}{x}$$

$$\boxed{10} \quad y' = xe^{y-x^2}$$